

**Topic 6**

# Trigonometry

**Concept guide · Worked examples · Cambridge-style practice · Fully explained answers**

**Learning focus**

Explain every method before practising. Use the original practice questions to check calculation, reasoning and communication.

**Topic map**

|            |                                     |
|------------|-------------------------------------|
| <b>6.1</b> | Pythagoras theorem                  |
| <b>6.2</b> | Trigonometric ratios                |
| <b>6.3</b> | Angles of elevation and bearings    |
| <b>6.4</b> | Sine and cosine rules               |
| <b>6.5</b> | Area of a non-right-angled triangle |
| <b>6.6</b> | Three-dimensional problems          |

Core foundations and Extended applications are integrated throughout. Study each example, then attempt the practice paper without the notes.

# 1. Right triangles

Pythagoras gives  $\text{hypotenuse}^2 = \text{other side}^2 + \text{other side}^2$ . The hypotenuse is opposite the right angle. For a missing shorter side, subtract squared lengths before taking a square root. In 3D, use one right triangle to find a face diagonal, then another for a space diagonal.

## Worked example

A 6-8-right triangle has hypotenuse  $\sqrt{6^2 + 8^2} = 10$ .

Exam check: Check the longest side is the hypotenuse before substituting.

# 2. Sine, cosine and tangent

For a marked acute angle in a right triangle,  $\text{sine} = \text{opposite}/\text{hypotenuse}$ ,  $\text{cosine} = \text{adjacent}/\text{hypotenuse}$ ,  $\text{tangent} = \text{opposite}/\text{adjacent}$ . Label sides relative to that angle before choosing. Use inverse trig for an angle and check the calculator is in degrees.

## Worked example

Opposite = 5 and adjacent = 12, so  $\theta = \tan^{-1}(5/12) = 22.6^\circ$  to one decimal place.

Exam check: Do not confuse an angle of elevation with the angle at the top of a vertical structure.

# 3. Non-right triangles

The sine rule links side/sine of its opposite angle for an opposite pair. The cosine rule  $c^2 = a^2 + b^2 - 2ab \cos C$  uses the included angle C. The area formula is  $\frac{1}{2} ab \sin C$  for two sides and the included angle. Sketch and label before selecting a formula.

## Worked example

Sides 6 and 8 contain  $30^\circ$ . Area =  $0.5 \times 6 \times 8 \times \sin 30^\circ = 12$  square units.

Exam check: With two sides and the included angle, cosine rule is usually the direct route to the third side.

# 4. Bearings, heights and 3D

Draw a plan view for bearings and a side elevation for height. In a cuboid, the face diagonal may form one side of a right triangle with vertical height. An angle between a line and a plane is measured against the line projection onto that plane.

## Worked example

A cuboid has base 3 cm by 4 cm and height 12 cm. Base diagonal = 5 cm; space diagonal =  $\sqrt{5^2 + 12^2} = 13$  cm.

Exam check: Show which triangle contains the required angle; a 3D sketch alone can hide the right angle.

## 5. Exact values and rounding

Recall  $\sin 30^\circ = 1/2$ ,  $\cos 60^\circ = 1/2$  and  $\tan 45^\circ = 1$ , plus the standard  $0^\circ$ ,  $30^\circ$ ,  $45^\circ$ ,  $60^\circ$ ,  $90^\circ$  values. Keep full calculator precision through intermediate stages, then round as directed. If a question requests an exact value, retain roots or  $\pi$  where appropriate.

### Worked example

In a right triangle with hypotenuse 10 and angle  $30^\circ$ , the opposite side is  $10 \sin 30^\circ = 5$ .

Exam check: An intermediate rounded angle can noticeably change a later length.

## Before the practice paper

### Method and accuracy

Write the formula or first operation, show the intermediate step, give units when needed and check by substitution, estimation or a sketch. On non-calculator work, keep exact forms when requested.



# Practice paper

## Topic 6 · Trigonometry

### Instructions

Original Cambridge-style practice. Answer all questions. Section A is multiple choice; Section B requires working and final answers. Try Section A without a calculator; use a calculator in Section B only where needed.

|                 |  |
|-----------------|--|
| Candidate name: |  |
| Class / date:   |  |

**Total: 33 marks**

## Section A · Multiple choice

Choose one answer for each question.

1. Hypotenuse of sides 5 and 12? [1]

- A 11
- B 12
- C 13
- D 17

2. For opposite 3, hypotenuse 5,  $\sin \theta$  is [1]

- A  $\frac{3}{5}$
- B  $\frac{4}{5}$
- C  $\frac{5}{3}$
- D  $\frac{3}{4}$

3. For adjacent 4, opposite 3,  $\tan \theta$  is [1]

- A  $\frac{4}{3}$
- B  $\frac{3}{4}$
- C  $\frac{3}{5}$
- D  $\frac{4}{5}$

4. Which is the sine rule? [1]

- A  $\frac{a}{\sin A} = \frac{b}{\sin B}$
- B  $a^2 + b^2 = c^2$  for all triangles
- C  $\sin A + \sin B = 1$
- D  $a \cos A = b \cos B$

5. Area given a, b and included C is [1]

- A  $ab \sin C$
- B  $ab/2$
- C  $ab \cos C/2$
- D  $ab \sin C/2$

6.  $\cos 60^\circ$  equals [1]

- A 0
- B  $\frac{1}{2}$
- C  $\frac{\sqrt{2}}{2}$
- D 1

7. If  $\tan \theta = 1$  for an acute angle,  $\theta$  is [1]

- A  $30^\circ$
- B  $45^\circ$
- C  $60^\circ$
- D  $90^\circ$

8. Cosine rule with included angle C gives  $c^2 =$  [1]

- A  $a^2 + b^2 + 2ab \cos C$
- B  $a^2 + b^2 - 2ab \cos C$
- C  $a^2 - b^2$
- D  $ab \cos C$

## Section B · Structured questions

9. A right triangle has legs 9 cm and 12 cm.

(a) Calculate the hypotenuse. [2]

(b) Find the sine of the angle opposite the 9 cm side. [2]

(c) Find the angle to one decimal place. [2]

**[Total: 6]**

10. A vertical tower casts a 20 m shadow. The Sun elevation is  $35^\circ$ .

(a) Write an equation involving the tower height  $h$ . [2]

(b) Calculate  $h$  to one decimal place. [2]

(c) Explain why tangent is the appropriate ratio. [2]

---

---

---

**[Total: 6]**

11. Triangle ABC has  $AB = 8$  cm,  $AC = 10$  cm and angle  $BAC = 60^\circ$ .

(a) Calculate BC using the cosine rule. [3]

(b) Calculate its area. [2]

(c) State the angle opposite BC. [1]

**[Total: 6]**

12. A cuboid is 6 cm long, 8 cm wide and 24 cm high.

(a) Find the base diagonal. [2]

(b) Find the space diagonal.

[2]

(c) Find the angle the space diagonal makes with the base, to one decimal place.

[3]

**[Total: 7]**



# Worked solutions

**Fold back this section while completing the practice paper.**

## Marking routine

Award method marks when the mathematical setup is correct even if an arithmetic slip follows.  
Check each part independently.

## Section A · Answers and reasons

1. C — 13. Pythagoras gives  $\sqrt{5^2+12^2} = 13$ .
2. A —  $3/5$ . Sine relates opposite to hypotenuse.
3. B —  $3/4$ . Tangent relates opposite to adjacent.
4. A —  $a/\sin A = b/\sin B$ . Each side is divided by sine of its opposite angle.
5. D —  $ab \sin C/2$ . Triangle area for two sides and included angle is  $ab \sin C/2$ .
6. B —  $1/2$ . The exact cosine value at  $60^\circ$  is  $1/2$ .
7. B —  $45^\circ$ . The acute angle with equal opposite and adjacent sides is  $45^\circ$ .
8. B —  $a^2+b^2-2ab \cos C$ . The included angle is subtracted in  $c^2 = a^2 + b^2 - 2ab \cos C$ .

## Section B • Worked solutions

9. a)  $\sqrt{9^2+12^2}=15$  cm [2]. b)  $9/15 = 3/5$  [2]. c)  $\sin^{-1}(3/5) = 36.9^\circ$  [2].

10. a)  $\tan 35^\circ = h/20$  [2]. b)  $h = 20\tan 35^\circ = 14.0$  m [2]. c) Opposite height and adjacent shadow are known; tangent relates them [2].

11. a)  $BC^2 = 8^2 + 10^2 - 2(8)(10)\cos 60^\circ = 84$ ;  $BC = \sqrt{84} \approx 9.17$  cm [3]. b)  $0.5(8)(10)\sin 60^\circ = 20\sqrt{3} \approx 34.6$  cm<sup>2</sup> [2]. c) Angle BAC [1].

12. a)  $\sqrt{6^2 + 8^2} = 10$  cm [2]. b)  $\sqrt{10^2 + 24^2} = 26$  cm [2]. c)  $\tan \theta = 24/10$ ,  $\theta = 67.4^\circ$  [3].

### Review your errors

Mark the first wrong step: setup, algebra, arithmetic, units, graph or reasoning. Redo that part without seeing the solution.

Syllabus reference: Cambridge IGCSE Mathematics 0580, 2025-2027,  
[www.cambridgeinternational.org/Images/662466-2025-2027-syllabus.pdf](https://www.cambridgeinternational.org/Images/662466-2025-2027-syllabus.pdf). Paper style informed by the June 2026 papers and  
examiner report provided by the user. All practice questions are original.